Data poisoning in intrusion systems

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Abstract— **The machine learning based (IDS) is used to detect the malicious network attack. Although the performance can get affected by the attackers who change the data that's used in the training set. In this project we will examine flipping labels to be as poisoned data and do the attack. And this is were labels will be changed on purpose to weaken the prediction of the model. In the experiment we will use a public cybersecurity dataset to train an intrusion detection system (ids) and then assess the effects by adding different poisoning levels 5/10/20. Later on, a defence technique will be added and evaluated to know if it can minimise the poisoned data impacts. At last we need to have an evaluation metrices to see the performance of the experiment: accuracy, precision, recall, f1 score, false positive rate. The main objective of the project is to have a practical knowledge of data poisoning risks and what defences can be implemented for preventing risks.**

Keywords—data poisoning, machine learning, intrusion detection system, label flipping, cybersecurity.

I. INTRODUCTION

1.1 Problem Statement

Machine learning (ML) is widely used in cybersecurity to identify suspicious network activities and detect potential cyberattacks and prevent them. Machine learning-based Intrusion Detection Systems (IDS) analyse network traffic and classify activities as normal or malicious. However, the performance and reliability of these systems depend on the quality and accuracy of the data used during training. A major security concern is data poisoning, which refers to when attackers intentionally manipulate the training data to affect a machine learning model's prediction. This project focuses on label–flipping attacks, a type of data poisoning in which the labels of training samples are changed. For example, malicious network traffic may be incorrectly labelled as normal traffic, causing the model to learn misleading patterns and potentially reducing its ability to detect cyberattacks. The main problem investigated in this project is how label-flipping data poisoning affects the performance of machine learning-based intrusion detection system and whether a simple defence method can reduce the impact caused.

1.2 Motivation

Machine learning-based intrusion detection systems play an important role in identifying potential cyber threats and supporting network security. However, these systems may become less reliable when attackers manipulate the data used during training. The incorrect training labels will influence the model's ability to differentiate between normal and malicious activities. The motivation behind this project is to understand how label-flipping attacks influence the intrusion detection performance and investigate a practical method for reducing their effects. By analysing different levels of data poisoning, the project aims to demonstrate the importance of reliable training data and explore different ways to improve the robustness of machine learning-based cybersecurity systems.

1.3 Project Scope

This project focuses on evaluating label-flipping data poisoning attacks against a machine learning-based network intrusion detection system (IDS). The project will use the publicly available cybersecurity dataset, which contains normal and

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malicious network traffic. And then if the is suitable for intrusion detection it will be selected. To simulate data poisoning, the labels of selected training samples will be intentionally changed at three different poisoning levels: 5%, 10%, and 20%. The model will be trained again using the modified datasets, and its performance will be compared with the original model. Additionally, the project will evaluate a simple defence strategy based on identifying potentially mislabelled training samples using cross-validation predictions will be evaluated. The effectiveness of the defence will be assessed using accuracy, precision, recall, F1-score, and false positive rate. The project will focus on binary classification and label-flipping attacks in a controlled testing environment. Other types of adversarial attacks, such as backdoor attacks and evasion attacks, will not be included.

1.4 Real-World Relevance

Machine learning-based intrusion detection systems play an important role in cybersecurity by detecting suspicious network activities and potential threats. Organizations use these systems to help protect their networks, information, and digital resources from cyberattacks. However, when attackers successfully manipulate the training data used by these systems, the resulting models may become less reliable and accurate. For example, if malicious network traffic is incorrectly labelled as normal, the machine learning model may struggle to recognize similar attacks in the future. This could lead to security threats going undetected and increase the risk of cyberattacks. This project addresses an important cybersecurity concern by investigating how changes to training labels affect the performance of intrusion detection system. By comparing the results before poisoning, after poisoning, and after applying a defence method, the project will help demonstrate the effects of label-flipping attacks and explore various ways to improve the security and reliability of machine learning-based intrusion detection systems.

1.5 research questions

Our key objective in the research is to analyse the effects of label flipping data poisoning attack on the Machine learning based intrusion detection (IDS) performance and if the defence method applied can minimise the effects. Therefore, our project will highlight the following questions for the research: Q1: how will the different label flipping data poisoning levels work on the ability of the Machine learning (IDS) system, to figure out the malicious network traffic? Q2: in which way will the effectiveness of a trained model that has poisoned data be compared with the original trained model? Q3: will the defence method used decrease the negative consequences of flipping labels attack and then enhance the (IDS) performance?

II. LITERATURE REVIEW

2.1 Machine Learning in IDS

In cybersecurity, machine learning (ML) has now become very important, especially when we want to identify some of the questionable network activities. Machine learning techniques are used by intrusion detection systems (IDS) to test the network data and categorize activity as malicious or normal. These tools can help us detect possible cyberattacks and find trends in the network data. Adversarial machine learning attacks against traditional intrusion detection systems were all tested by Alotaibi and Rassam [1]. Regarding to their research, machine learning methods can enhance intrusion detection by finding patterns which are connected to malevolent activity. The researchers do also say that these systems are still susceptible to adversarial assaults that alter data while they are being tested or trained as well. These attacks have the potential to reduce the detection accuracy and increase the likelihood of wrong predictions. Similar to this, studies on

adversarial machine learning for network intrusion detection have showed that the quality and security of the data are very important to model performance to the learning algorithm. Which makes it important to look into how intrusion detection systems can be protected from the threats and how attackers can alter training datasets. Moreover, Buczak and Guven [2] has examined the machine learning techniques for the (IDS) to tell the importance of choosing suitable datasets and methods. Likewise, Khraisat et al. [3] also analysed multiple (IDS) methods and demonstrated the issues with attacks and dataset.

2.2 Data Poisoning

An adversarial machine learning attack known as "data poisoning" happens when an attacker modifies training data on purpose to affect the model's learning process. Data poisoning targets the data used to create the model, as opposed to attacks that happen the after training. After that immediately the model might pick up erroneous trends and create forecasts which are not accurate. Label flipping is a popular tactic where an attacker modifies the labels which are given to training samples without necessarily altering the original data attributes. For example, network traffic that indicates of a cyberattack could be classified as typical activity by mistake. The model's capacity to discriminate between legitimate and malicious communications may be affected if it is trained with erroneous labels. Label-flipping attacks against federated learning-based intrusion detection systems were tested by Lavaur, Busnel, and Autrel [2]. They looked at how collaborative intrusion detection models were impacted by malicious modifications to training labels. The impact of label-flipping attacks were different based on the model's design and dataset features, the researchers discovered. Their results showed, although their efficacy are not all the same depending on the experimental settings, label-flipping attacks can have a substantial impact on detection performance. Because both studies check how altered training labels affect intrusion detection, they are very pertinent to the proposed endeavour. On the other hand, Lavaur et al.[4] concentrated on federated learning, where a shared model is trained by multiple participants. Instead, the proposed experiment will test the consequences of label flipping using a more straightforward, centralized machine learning model. Goldblum et al. [5] discussed the poisoned data attacks by explaining how changed trained sets affect Ml system. where Biggio et al. [8] has studied poisoned attacks on svm and demonstrated that modified training data sets can have an outcome of more errors while classifying so eventually Ml systems can be compromised when the training data is not accurate. Nest [9] has also addressed multiple machine Learning attacks and highlighted the need of awareness to minimize the risks.

2.3 Defense Methods

Researchers have suggested many methods to try to reduce the effects of data poisoning attacks on machine learning systems. These methods contain recognizing questionable training examples, fixing inaccurate labels, and eliminating possibly contaminated data prior to model training. Monkam, De Lucia, and Bastian [3] suggested a technique for identifying data poisoning attacks on network intrusion detection systems that depend on machine learning. Their method integrates topological data analysis with unsupervised clustering to detect unusual patterns in training data. The researchers showed how contaminated samples could be grouped into clusters and eliminated prior to training the intrusion detection model. This study emphasizes the significance of detecting contaminated data prior to its impact on the learning process. The suggested method entails sophisticated techniques, potentially raising the implementation complexity in comparison to more straightforward data-cleaning methods. Taheri et al. [7] studied

label-flipping attacks on malware detection systems that are based on machine learning. Their research suggested semi-supervised defence techniques to detect and amend altered labels. The researchers assessed their methods with Android malware datasets and noted enhancements in classification accuracy compared to a previously established approach. While Taheri et al.[7] concentrated on malware detection instead of network intrusion detection, this shows that recognizing altered labels could help in mitigating the impacts of data poisoning. This shows important context for choosing a defence strategy for the suggested project. Overall, these studies shows that identifying training samples and enhancing label quality are viable strategies for protecting machine learning systems from data poisoning. Nonetheless, the efficacy of each approach relieson the dataset, attributes of the attack, and the machine learning model employed.

2.4 Comparison

Earlier studies have explored adversarial assaults on machine learning systems and suggested multiple defensive approaches. Nonetheless, the research varies in its goals, techniques, and cybersecurity implementations. Alotaibi and Rassam [1] offered an extensive overview of adversarial machine learning threats and countermeasures in intrusion detection systems. Their study helps us in understanding the different security risks that will affect Machine learning based intrusion detection however, it does not include a thorough experimental evaluation of label-flipping attacks. Lavaur et al. [4] focused on label-flipping attacks within federated intrusion detection systems. Their study gives experimental support which indicates that altered training labels can influence detection performance. Their research emphasizes collaborative federated learning instead of a traditional centralized model. Monkam et al. [6] created a technique for identifying contaminated training samples by employing topological data analysis and clustering. Their method helps in detecting the contaminated data prior to training, whereas Taheri et al. [7] explored techniques for rectifying altered labels in malware detection systems. Steinhardt et al. [10] researched into poisoned data defences their research has showed a dependable relatonship between model performance and dataset. Similarly like Northcutt et al. [11], he has established a technique to recognise labeling errors.

2.5 Research Gap

While earlier research has looked into label-flipping attacks and defences against the data poisoning, most of it centres on elaborate methods, such as federated learning, deep learning, and sophisticated data analysis strategies. The suggested project will concentrate on a simpler and more manageable experimental environment. It will test how varying percentages of label-flipping data poisoning influence a centralized machine learning-driven intrusion detection system. The project will additionally assess a straightforward defence approach to check whether it can effect altered training labels. Our goal is not to present a totally novel defence algorithm, but to offer a distinct and replicable evaluation of model performance regarding to poisoning, following poisoning, and after implementing a defence. This method will help in illustrating the impact of data poisoning on intrusion detection effectiveness and offer a practical assessment of a defence strategy with a simple machine learning context.

III. Methodology

3.1 proposed approach

The idea of this project uses an experimental approach to study the effect of the label flipping data poisoning attack on machine learning it is based : ids (intrusion detection system). Moreover, our experiment contains training a machine learning model in addition of a cybersecurity

dataset for normal and malicious network traffic. In the first step, training the model will be only with normal source of the dataset to examine and evaluate the dataset performance normally. Following that, we will mimic the label flipping attack by flipping an amount of training labels in purpose. The model will be trained again but with the poisoned dataset and test it this will show how it changed and compare both testings, we will also see how the changes that we did will impact on its ability to identify a cyber attack. Lastly, to have a full evaluation we need to implement a basic defence technique that will be evaluated, this will show whether it is able to reduce the data poisoning effects or not. The evaluation of defence success, we will compare the results of normal, poisoned, defended models.

3.2 dataset and tools

We will be using a public dataset in our project on cybersecurity with normal and malicious network traffic. To check if the dataset is suitable or fits in will be by comparing with intrusion detection relevance, the data quality, ease of implementing. Then the final dataset will be selected. Furthermore, the dataset will be processed to be ready for the binary classification where there will be two classes for the network traffic :

- Benign traffic (normal)
- Attack (malicious traffic)

As well as the implementation that will use python and scikit-learn for machine learning, pandas for data preparation, matplotlib to present the outcomes, google colab will also be used to run the model, while github for supporting the collaboration of the project.

3.3 experiment procedure

There are four main stages that we will follow in this experiment:

First stage: baseline model

To begin with the first stage, we will choose a machine learning dataset, then it must be prepared and go through preprocessing stage. So, the preparation will handle missing values, selection of features, data separation to train and test The ML model. Therefore, the model will be trained by using the normal dataset that contains original labelling. The performance will be evaluated with the original test data. Additionally, the results will give us a starting point for compare with the experiment later.

Second stage: label flip, poisoning attack

In this stage, we will flip labels on purpose and change some amounts of the labels in the training dataset. Poisoning level will be : 5%, 10% and 20%. In each of these poisoning levels the labels in the training set will be flipped from the original class labelling to the opposite class. So, during experiment planning the final poisoning percent would be confirmed, as well as it will help with comparing the outcomes fairly.

Third stage: defence method

Stage three will attempt to minimize poisoned training data effects, so we will go over the research and figure out the strategy that will be more feasible to apply and then choose a defence technique that is simple. Moreover, there is one approach for defending and it is by comparing the labels of training samples with predictions of Ml model, in order to find training samples that may contain incorrect labels. Next, samples that contains unusual, inconsistence labelling can be examined, deleted, this will happen before training the model again. After all, not every wrong or strange sample have to be poisoned. So eventually, the effectiveness of defence method needs to be evaluated carefully in order to check if there will be any improvement while detecting efficiency without removing a lot of data samples that may be useful. Lastly in this stage, the first testing data will be used to defend the model.

Fourth stage: comparison

In the last stage, where the comparison of the first original results and the poisoned models as well as the model that has been trained after adding the defence technique will happen. However, there is two main questions to be considered and they are: at what extinct will flipping the labels impact on the efficiency of the model to recognize attacks? And will the defence method that is chosen be able to reduce the negative impacts of poisoned training samples? Therefore, the outcomes are going to be presented in tables and graphs so it will be easy to see the differences in result and experiment.

3.4 evaluation metrices

There will be multiple measurements to evaluate the model performance. Accuracy will measure total percentage of the predictions that are correct. Precision is the behaviours that is classified as an attack and they are true ones. Recall will measure true attacks that the model detected. F1 score is a combination of precision and recall. At last, false positive rates will be the rate of mistakenly identified a normal network traffic as being an attack. However, measuring metrices are crucial for the model since the model can be accurate yet still fail to detect some attacks. Hence, the focus will be on recall and the f1 score at the comparison stage for the results. As well as we will use the same test sample on the whole experiment to ensure that the variations in results are because of changing training data not the testing data. The importance of using performance metrices to compare between original and poisoned results will be discussed. Sokolova and Lapalme [12] has done multiple performance measuring and discussed the importance of using them.

3.5 ethical considerations

This project is being implemented in a safe setting for academic and research purpose. Our group will use the public cybersecurity dataset and will not carry out any attacks on a real network. Also, the poisoning attacks will be as simple as changing the labels in a local copy of the training sample. There will be no illegal access, interruptions, malicious activity. As well as, before choosing the dataset we will investigate the usage policy and condition and then we will start using it. Following that, the software tools, research's, dataset that is used in the project will also be used and cited correctly. Since the key objective of the project is to gain more knowledge and understand data poisoning risks and how machine learning detection system can be made more effective.

IV. Team Plan

This project will be done by three group members. And the responsibilities will be divided by all three of us to make sure that research, testing, implementing, documentation are done effectively. All three members will engage in evaluating the results and be ready to present the work.

Week4	Deliverable 1, review research paper, GitHub repository
Week5	Selection and preparation of the cybersecurity dataset; developing the model
Week6	Simulating label flipping attack at 5,10,20 levels
Week7	Implementation of defence technique, comparing performance of model
Week8	Finale analysing of all test results, documentations, presentation

References

[1] A. Alotaibi and M. A. Rassam, "Adversarial Machine Learning Attacks against Intrusion Detection Systems: A Survey on Strategies and Defense," *Future Internet*, vol. 15, no. 2, Art. no. 62, 2023. https://doi.org/10.3390/fi15020062

[2] A. L. Buczak and E. Guven, "A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection," *IEEE*

Communications Surveys & Tutorials, vol. 18, no. 2, pp. 1153–1176, 2016. https://doi.org/10.1109/COMST.2015.2494502

[3] A. Khraisat, I. Gondal, P. Vamplew, and J. Kamruzzaman, "Survey of intrusion detection systems: techniques, datasets and challenges," *Cybersecurity*, vol. 2, Art. no. 20, 2019. https://doi.org/10.1186/s42400-019-0038-7

[4] L. Lavaur, Y. Busnel, and F. Autrel, "Investigating the Impact of Label-flipping Attacks against Federated Learning for Collaborative Intrusion Detection," *Computers & Security*, vol. 156, Art. no. 104462, 2025. https://doi.org/10.1016/j.cose.2025.104462

[5] M. Goldblum et al., "Dataset Security for Machine Learning: Data Poisoning, Backdoor Attacks, and Defenses," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 2, pp. 1563–1580, 2023. https://doi.org/10.1109/TPAMI.2022.3162397

[6] G. F. Monkam, M. J. De Lucia, and N. D. Bastian, "A Topological Data Analysis Approach for Detecting Data Poisoning Attacks Against Machine Learning Based Network Intrusion Detection Systems," *Computers & Security*, vol. 144, Art. no. 103929, 2024. https://doi.org/10.1016/j.cose.2024.103929

[7] R. Taheri, R. Javidan, M. Shojafar, Z. Pooranian, A. Miri, and M. Conti, "On Defending Against Label Flipping Attacks on Malware Detection Systems," *Neural Computing and Applications*, vol. 32, no. 18, pp. 14781–14800, 2020. https://doi.org/10.1007/s00521-020-04831-9

[8] B. Biggio, B. Nelson, and P. Laskov, "Poisoning Attacks against Support Vector Machines," in *Proceedings of the 29th International Conference on Machine Learning (ICML)*, 2012. https://icml.cc/2012/papers/880.pdf

[9] A. Vassilev, A. Oprea, A. Fordyce, H. Anderson, X. Davies, and M. Hamin, *Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations*, NIST AI 100-2e2025, National Institute of Standards and Technology, 2025. https://doi.org/10.6028/NIST.AI.100-2e2025

[10] J. Steinhardt, P. W. Koh, and P. Liang, "Certified Defenses for Data Poisoning Attacks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. https://proceedings.neurips.cc/paper_files/paper/2017/hash/9d7311ba459f9e45ed746755a32dcd11-Abstract.html

[11] C. G. Northcutt, L. Jiang, and I. L. Chuang, "Confident Learning: Estimating Uncertainty in Dataset Labels," *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021. https://doi.org/10.1613/JAIR.1.12125

[12] M. Sokolova and G. Lapalme, "A Systematic Analysis of Performance Measures for Classification Tasks," *Information Processing & Management*, vol. 45, no. 4, pp. 427–437, 2009. https://doi.org/10.1016/j.ipm.2009.03.002